# DREAM6-ZR Boundary Closure Checkpoint

Exact graded entry contraction after the six-dimensional phase reduction

3 September 2026

**Abstract**

This checkpoint records the current exact status of the DREAM6 zero-repair boundary program after the six-dimensional local phase reduction and the audited v0.2 positive-phase experiment.

A new exact result is proved here. Let

$$B(t) = \sum_{j \geq 0} b_j t^j$$

be a boundary buffer law with $b_1 = 0$ and $0 \leq b_0 \leq 1/2$. There is a causal transport for which the critical $x = 0$ child enters the previously identified CORE class, the $x = 4$ child enters CORE or the terminal state $t$, and only the $x = 2$ child carries a transient. Its new constant mass satisfies

$$b_0' \leq \tau(b_0) := \max\left\{0, \frac{48b_0 - 9}{38}\right\}, \qquad b_1' = 0.$$

Starting from the hard-edge product boundary value $b_0 = 1/2$, two such peeling steps give

$$\frac{1}{2} \longmapsto \frac{15}{38} \longmapsto \frac{189}{722} < \frac{13}{48}.$$

At $b_0 \leq 13/48$, a final landing transport sends the $x = 2$ child directly into CORE. Thus the critical-channel boundary transient has length at most three.

The construction also pays the exact local repair debt $1/32$ with a strict margin

$$\delta(b_0) = \frac{23}{128} - \frac{b_0}{4},$$

whose worst entry value is $7/128$.

This closes the local hard-edge entry problem on the critical channels. It does *not* yet prove the full causal addition lemma or the $O(\log n)$ theorem: the $x = 1, 3$ recursive obligations, the depth/fuel semantics of the terminal state $t$, and the ordered-history ("upstairs") phase lift remain separate gates.

## 1 Status ledger

The exact coarse block is

$$P_{\mathrm{b}} = \frac{1}{64}(1, 12, 38, 12, 1), \qquad Q_{\mathrm{b}} = \frac{1}{64}(0, 16, 32, 16, 0),$$

with primitive defect

$$\boxed{P_{\mathrm{b}}(t) - Q_{\mathrm{b}}(t) = \frac{(1-t)^4}{64}}.$$

At $r = 0$, the natural interior continuation exposes the two illegal cells

$$(2,1), \qquad (4,3),$$

with masses

$$\frac{3}{128}, \qquad \frac{1}{128},$$

hence total local debt

$$\boxed{\frac{1}{32}}.$$

The preceding phase-compression note established, for the natural nine-coordinate local boundary readout,

$$\boxed{\dim \mathcal{R}_m(\ker M_m) = 6 \qquad (m \geq 2).}$$

One convenient phase basis is

$$\Phi = (H_3(B_0),\ H_1(B_2),\ H_3(B_2),\ H_1(B_4),\ H_3(B_4),\ D_{\mathrm{sh}}).$$

The v0.2 compiler then tested positivity and one-step recursive repairability on a finite semantic policy graph. Its completed run contained 119 unique laws, 100 LP-feasible nodes and 19 infeasible nodes. Of the 100 feasible nodes, 99 had positive common repair margin. No unknown or numerical-reject status occurred. The worst equality residual was $6.25 \times 10^{-12}$, with zero inequality violation, zero clipped negative transport mass, and minimum transport cell 0.

These data motivated the exact reduction below, but none of the proofs in this checkpoint use the numerical policy graph.

## 2 Transport notation

Let

$$B(t) = \sum_{j \geq 0} b_j t^j, \qquad b_j \geq 0, \qquad \sum_j b_j = 1,$$

and write

$$q_u = (Q_{\mathrm{b}} * B)_u.$$

A local causal transport is a nonnegative array $\pi(x, u)$ satisfying

$$\sum_u \pi(x,u) = P_{\mathrm{b}}(x),$$

$$\sum_x \pi(x,u) = q_u,$$

and

$$\pi(x,u) = 0 \qquad (u < x).$$

The residual child law on row $x$ is

$$B_x(j) = \frac{\pi(x, x+j)}{P_{\mathrm{b}}(x)}.$$

The two parent-determined donor capacities are

$$D_{21}(B) = \frac{13}{64} - \frac{1}{4} b_0,$$

and

$$D_{43}(B) = \frac{63}{64} - \left(b_0 + \frac{3}{4} b_1 + \frac{1}{4} b_2\right).$$

# 3 CORE and terminal sorts

Define

$$\boxed{\mathcal{C} := \left\{B : b_0 = 0, \quad b_1 \le \frac{13}{16}\right\}.} \tag{1}$$

The distinguished terminal law is

$$\boxed{\mathcal{T} := \{t\}.}$$

The threshold 13/16 is a Hall-capacity threshold, not a fitted constant. Indeed, if $b_0 = 0$, then

$$q_2 = \frac{1}{4} b_1.$$

For the $x = 2$ child to have zero constant coefficient one needs $\pi(2, 2) = 0$, so column 2 must be absorbed by rows $0, 1$, whose total capacity is

$$P_{\mathrm{b}}(0) + P_{\mathrm{b}}(1) = \frac{13}{64}.$$

Hence the sharp condition

$$\frac{1}{4} b_1 \le \frac{13}{64} \quad \Longleftrightarrow \quad b_1 \le \frac{13}{16}.$$

**Proposition 1** (Critical-channel CORE closure)**.** *For every $B \in \mathcal{C}$, there exists a causal transport such that*

$$B_0 \in \mathcal{C}, \qquad B_2 \in \mathcal{C}, \qquad B_4 \in \mathcal{C} \cup \mathcal{T}.$$

*Moreover the local repair screen can attain the maximal margin*

$$\boxed{\delta_{\mathrm{core}} = \frac{23}{128}.}$$

*Proof.* Since $b_0 = 0$,

$$q_0 = q_1 = 0, \qquad q_2 = \frac{1}{4} b_1 \le \frac{13}{64}.$$

Thus column 2 can be assigned entirely to rows $0, 1$, so $\pi(2, 2) = 0$. This gives $(B_2)_0 = 0$.

Also

$$q_3 = \frac{1}{2} b_1 + \frac{1}{4} b_2 \le \frac{1}{4} + \frac{1}{4} b_1 \le \frac{29}{64}.$$

Therefore

$$(B_2)_1 = \frac{\pi(2, 3)}{38/64} \le \frac{29}{38} < \frac{13}{16},$$

so $B_2 \in \mathcal{C}$. The $x = 0$ child has zero first two coefficients because $q_0 = q_1 = 0$, hence $B_0 \in \mathcal{C}$.

Finally,

$$F_q(4) = b_1 + \frac{3}{4} b_2 + \frac{1}{4} b_3 \le \frac{3}{4} + \frac{1}{4} b_1 \le \frac{61}{64}.$$

Thus at least 3/64 of the column mass lies in $u \ge 5$. If $q_5 \ge 1/64$, put row 4 entirely at $u = 5$, giving $B_4 = t$. Otherwise more than 2/64 lies in $u \ge 6$, enough to place the entire row 4 there; then $(B_4)_0 = (B_4)_1 = 0$, so $B_4 \in \mathcal{C}$.

For the repair margin,

$$D_{21} = \frac{13}{64},$$

so the first debt leaves

$$D_{21} - \frac{3}{128} = \frac{23}{128}.$$

The other repair inequalities have strictly larger slack on $\mathcal{C}$, so this upper bound is attained. $\square$

## 4 The graded entry classes

The new observation is that the hard boundary transient collapses to one scalar.

For $0 \le c \le 1/2$, define

$$\mathcal{E}(c) := \{B : b_1 = 0, \quad 0 \le b_0 \le c\}.$$

Define the peeling map

$$\boxed{\tau(c) = \max\left\{0, \frac{48c - 9}{38}\right\}.} \tag{2}$$

**Theorem 1** (Exact peeling step)**.** *Let $B \in \mathcal{E}(1/2)$, and put $c = b_0$. There exists a causal transport such that*

$$B_0 \in \mathcal{C}, \qquad B_4 \in \mathcal{C} \cup \mathcal{T},$$

*while the only critical transient satisfies*

$$(B_2)_1 = 0, \qquad (B_2)_0 \le \tau(c).$$

*Equivalently,*

$$B_2 \in \mathcal{E}(\tau(c)).$$

*The same transport pays the local boundary debt with margin*

$$\boxed{\delta(c) = \frac{23}{128} - \frac{c}{4} > 0.} \tag{3}$$

*Proof.* Because $b_1 = 0$,

$$q_0 = 0, \qquad q_1 = \frac{c}{4}, \qquad q_2 = \frac{c}{2}, \qquad q_3 = \frac{c}{4} + \frac{1}{4}b_2.$$

We construct the low-column part first. We forbid row 0 from column 1, which will force the $x = 0$ child into CORE, and we forbid row 2 from column 3, which will force $(B_2)_1 = 0$.

Put

$$r_2 := \left(q_1 + q_2 - \frac{13}{64}\right)_+,$$
$$r_3 := \left(q_1 + q_2 + q_3 - \frac{25}{64}\right)_+,$$

and

$$r := \max\{r_2, r_3\}.$$

We assign mass $r$ to the cell $(2, 2)$.

The first lower bound $r \ge r_2$ is exactly the condition that columns $1, 2$, after the row-2 contribution $r$, fit into rows $0, 1$. The second lower bound $r \ge r_3$ is exactly the condition that columns $1, 2, 3$, after the same contribution, fit into rows $0, 1, 3$. Since

$$q_1 = \frac{c}{4} \le \frac{1}{8} < \frac{12}{64},$$

column 1 itself fits in row 1, so these cumulative conditions are sufficient for the low-column allocation.

Now $b_2 \le 1 - c$, hence

$$r_3 \le \left(c + \frac{1-c}{4} - \frac{25}{64}\right)_+ = \left(\frac{48c - 9}{64}\right)_+,$$

while
$$r_2 = \left(\frac{48c-13}{64}\right)_+ .$$
Therefore
$$r \le \left(\frac{48c-9}{64}\right)_+ .$$
For $c \le 1/2$, this is at most $q_2 = c/2$, so the assignment to $(2,2)$ is available.

Consequently
$$(B_2)_0 = \frac{r}{38/64} \le \max\left\{0, \frac{48c-9}{38}\right\} = \tau(c),$$
and, by construction,
$$(B_2)_1 = 0.$$

The $x = 0$ row never uses columns $0, 1$, hence its first two residual coefficients vanish and $B_0 \in \mathcal{C}$.

For row 4, observe
$$F_q(4) = c + \frac{3}{4}b_2 + \frac{1}{4}b_3 \le c + \frac{3}{4}(1-c) = \frac{3}{4} + \frac{c}{4} \le \frac{7}{8}.$$
Thus at least $1/8$ of all column mass lies in $u \ge 5$. If $q_5 \ge 1/64$, place row 4 at $u = 5$, giving $t$. Otherwise more than $7/64$ lies in $u \ge 6$, so row 4 can be placed entirely there, giving a CORE child. Once the low columns and row 4 are fixed, all remaining rows $0, 1, 2, 3$ may use every remaining column $u \ge 4$, so the transport completes.

For the repair margin,
$$D_{21} = \frac{13}{64} - \frac{c}{4},$$
hence
$$D_{21} - \frac{3}{128} = \frac{23}{128} - \frac{c}{4} = \delta(c).$$
Furthermore
$$H_3(B) = c + \frac{1}{4}b_2 \le \frac{1}{4} + \frac{3}{4}c \le \frac{5}{8},$$
so
$$D_{43} \ge \frac{23}{64},$$
and therefore the $D_{43}$ repair slack is at least $45/128$. Finally $D_{\mathrm{sh}} \le 13/64$, giving union slack at least
$$\frac{21}{64} - \frac{c}{4},$$
which exceeds $\delta(c)$ by $19/128$. Thus the first donor inequality is the unique bottleneck and (3) is attained. $\square$

## 5 Landing into CORE

**Lemma 1** (Landing threshold)**.** *Let*
$$B \in \mathcal{E}\left(\frac{13}{48}\right).$$
*Then there is a causal transport for which*
$$B_0 \in \mathcal{C}, \qquad B_2 \in \mathcal{C}, \qquad B_4 \in \mathcal{C} \cup \mathcal{T}.$$

*Proof.* Again $b_1 = 0$. The total mass in columns $1, 2$ is

$$q_1 + q_2 = \frac{3}{4}c \leq \frac{13}{64} = P_{\mathrm{b}}(0) + P_{\mathrm{b}}(1).$$

Hence rows $0, 1$ can absorb columns $1, 2$, with row 0 avoiding column 1. Therefore

$$\pi(2, 2) = 0,$$

so $(B_2)_0 = 0$.

Moreover

$$q_3 = \frac{c}{4} + \frac{1}{4}b_2 \leq \frac{1}{4},$$

and therefore, regardless of how much of column 3 row 2 receives,

$$(B_2)_1 \leq \frac{1/4}{38/64} = \frac{8}{19} < \frac{13}{16}.$$

Thus $B_2 \in \mathcal{C}$. The $x = 0$ and $x = 4$ conclusions follow exactly as in theorem 1. $\square$

# 6 Finite transient from a hard-edge product boundary

The boundary grammar state has the form

$$G_0(A, C) = \frac{1}{2} + \frac{1}{2}A(t)C(t).$$

If

$$A(0) = C(0) = 0,$$

then

$$[AC]_{t^0} = 0, \qquad [AC]_{t^1} = 0,$$

and hence

$$\boxed{[G_0(A, C)]_{t^0} = \frac{1}{2}, \qquad [G_0(A, C)]_{t^1} = 0.}$$

Thus every such hard-edge product boundary state belongs to $\mathcal{E}(1/2)$.

Now

$$c_0 = \frac{1}{2}.$$

The first peeling step gives

$$c_1 = \tau(c_0) = \frac{15}{38}.$$

The second gives

$$c_2 = \tau(c_1) = \frac{189}{722}.$$

A direct comparison yields

$$189 \cdot 48 = 9072 < 9386 = 13 \cdot 722,$$

so

$$\boxed{\frac{189}{722} < \frac{13}{48}.}$$

Therefore the third critical $x = 2$ step lands in CORE by lemma 1.

**Corollary 1** (Three-step critical entry). *For every hard-edge product boundary state*

$$G_0(A, C), \qquad A(0) = C(0) = 0,$$

*there exists a graded critical-channel transport grammar of length at most three:*

$$\mathcal{E}\Big(\frac{1}{2}\Big) \longrightarrow \mathcal{E}\Big(\frac{15}{38}\Big) \longrightarrow \mathcal{E}\Big(\frac{189}{722}\Big) \longrightarrow \mathcal{C},$$

*where at every stage the $x = 0$ child enters CORE, the $x = 4$ child enters CORE or $t$, and only the $x = 2$ child continues the transient.*

The corresponding worst-case repair margins are

$$\delta\Big(\frac{1}{2}\Big) = \frac{7}{128},$$

$$\delta\Big(\frac{15}{38}\Big) = \frac{197}{2432},$$

$$\delta\Big(\frac{189}{722}\Big) = \frac{5279}{46208},$$

followed by the CORE margin

$$\frac{23}{128}.$$

In particular, the entire entry transient remains strictly inside the local positive repair region.

# 7 Finite critical grammar

The exact local grammar can therefore be written as

$$\boxed{\begin{array}{lcl} \mathcal{E}_2 & \xrightarrow{x=2} & \mathcal{E}_1, \\ \mathcal{E}_1 & \xrightarrow{x=2} & \mathcal{E}_0, \\ \mathcal{E}_0 & \xrightarrow{x=2} & \mathcal{C}, \end{array}}$$

with

$$\mathcal{E}_2 = \mathcal{E}(1/2), \qquad \mathcal{E}_1 = \mathcal{E}(15/38), \qquad \mathcal{E}_0 = \mathcal{E}(189/722),$$

and, at every one of these stages,

$$x = 0 : \quad \mathcal{C},$$

$$x = 4 : \quad \mathcal{C} \cup \mathcal{T}.$$

Once in CORE,

$$x = 0, 2 : \quad \mathcal{C},$$

$$x = 4 : \quad \mathcal{C} \cup \mathcal{T}.$$

This is the precise form of the "finite transient → invariant face" pattern that was only visible numerically in DREAM6-ZR-PHASE v0.2.

# 8 Independent computational crash-test

A separate verifier, `DREAM6_ZR_ENTRY_CERT_v01.py`, encodes the constrained transportation problems independently of the proof.

Its final regression run used:

- 310 extreme one-atom-tail contract tests;
- 10,000 random finite-support laws with $b_1 = 0$;
- tail support through index 32;
- explicit HiGHS primal and dual feasibility tolerance $10^{-9}$.

The result was

$$\boxed{\texttt{PASS}}$$

with no failed contract, maximum equality residual

$$4.44 \times 10^{-16},$$

and maximum inequality violation 0.

An earlier run with the solver default feasibility settings produced one apparent audit failure at equality residual $6.3 \times 10^{-8}$; tightening the solver feasibility tolerance removed it. This is recorded deliberately: the verifier treats numerical residuals as numerical evidence only and never as part of the exact proof.

# 9 What is now solved

The following statements are exact within the stated local transport model.

1. The primitive defect is $(1-t)^4/64$.
2. The local $r = 0$ debt is exactly $1/32$.
3. The natural local boundary readout has a six-dimensional support-independent latent phase.
4. The critical-channel CORE class

$$\mathcal{C} = \{b_0 = 0,\ b_1 \leq 13/16\}$$

   is positively closed up to the distinguished terminal sort $t$.
5. Every hard-edge product boundary state

$$G_0(A, C), \qquad A(0) = C(0) = 0,$$

   enters CORE after at most three $x = 2$ transient steps.
6. The entire entry path has a strictly positive exact repair margin, with worst value $7/128$.

Thus the *local hard-edge entry obstruction on the critical channels* is no longer an open numerical pattern. It has an explicit finite graded transport proof.

# 10 What is not yet solved

This checkpoint is intentionally not labeled “boundary closure proved.”

Three gates remain.

### 1. The $x = 1, 3$ recursive obligations

The phase compiler and the exact grammar above isolate the critical channels $x \in \{0, 2, 4\}$. A full $K_m$-type causal witness has five children. The $x = 1, 3$ branches must be incorporated into a well-founded induction or an exact relation calculus. In the original product grammar these branches are expected to correspond to same-boundary-form recursive calls with reduced factor depth, but this bookkeeping is not proved by the present state-only argument.

### 2. Terminal fuel

The law $t$ is a genuine base/terminal object, not an indefinitely viable state. Therefore a transition into $t$ is valid only when accompanied by the correct remaining-depth or fuel semantics. The local grammar identifies where $t$ can appear; it does not yet prove that every such appearance occurs at an admissible terminal depth.

### 3. Ordered-history lift

The coarse transport equations forget ordered-history phase information. The final causal construction must still lift the coarse graded grammar to the history-indexed setting without losing positivity. This remains the "upstairs" gate.

Consequently neither the full addition lemma nor the final $O(\log n)$ upper bound is claimed here.

## 11 Next exact target

The next object should not be a larger LP.

The natural target is a depth-labelled graded relation

$$\mathfrak{G}^{(r)}_{m,n}$$

whose state contains:

1. the interior buffer grade $r$;
2. the finite boundary grade $\mathcal{E}_2, \mathcal{E}_1, \mathcal{E}_0, \mathcal{C}, \mathcal{T}$;
3. the remaining factor-depth/fuel labels;
4. only the phase information needed for the ordered-history lift.

The desired theorem has the schematic form

$$\boxed{\mathfrak{G}^{(r)}_{m,n} \longrightarrow \{\mathfrak{G}^{(r')}_{m',n'}\}}$$

with a strictly decreasing lexicographic measure, for example

$$(m + n,\ \text{boundary grade},\ r),$$

and with $t$ admitted only at measure zero/base depth.

If this well-founded five-branch relation closes, the remaining local boundary problem is finished in the sense required by the dyadic Kovačević recursion. Only then should the mean recursion be promoted to the final logarithmic bound.

## 12 Checkpoint conclusion

The state of the program is now:

$$\boxed{\begin{array}{c}\text{quartic defect}\\ \Downarrow \\ \text{exact } 1/32 \text{ local debt}\\ \Downarrow \\ \text{six-dimensional local phase}\\ \Downarrow \\ \text{exact CORE face}\\ \Downarrow \\ \text{exact three-step hard-edge entry}\\ \Downarrow \\ \boxed{\text{remaining: five-branch depth-labelled causal lift}}\end{array}}$$

The important change is qualitative. The boundary is no longer represented as a large witness tree or a numerical phase cloud. On the critical channels, it has collapsed to a finite graded grammar with one scalar transient and explicit rational thresholds.